Dell XPS 13 9345 – EC Driver Reverse Engineering, 2026.10.24
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Introduction

This Qualcomm X1E-based laptop has 7 thermistors placed around the motherboard. Windows driver initializes the following 7 ADC channels:

- 0x4a, aka `ADC5_GEN3_AMUX1_GPIO_100K_PU(1)` corresponding to `SYS_THERM1`
- 0x4b, aka `ADC5_GEN3_AMUX1_GPIO_100K_PU(1)` corresponding to `SYS_THERM2`
- 0x44, aka `ADC5_GEN3_AMUX1_THM_100K_PU(1)` corresponding to `SYS_THERM3`
- 0x45, aka `ADC5_GEN3_AMUX2_THM_100K_PU(1)` corresponding to `SYS_THERM4`
- 0x46, aka `ADC5_GEN3_AMUX3_THM_100K_PU(1)` corresponding to `SYS_THERM5`
- 0x47, aka `ADC5_GEN3_AMUX4_THM_100K_PU(1)` corresponding to `SYS_THERM6`
- 0x48, aka `ADC5_GEN3_AMUX5_THM_100K_PU(1)` corresponding to `SYS_THERM7`

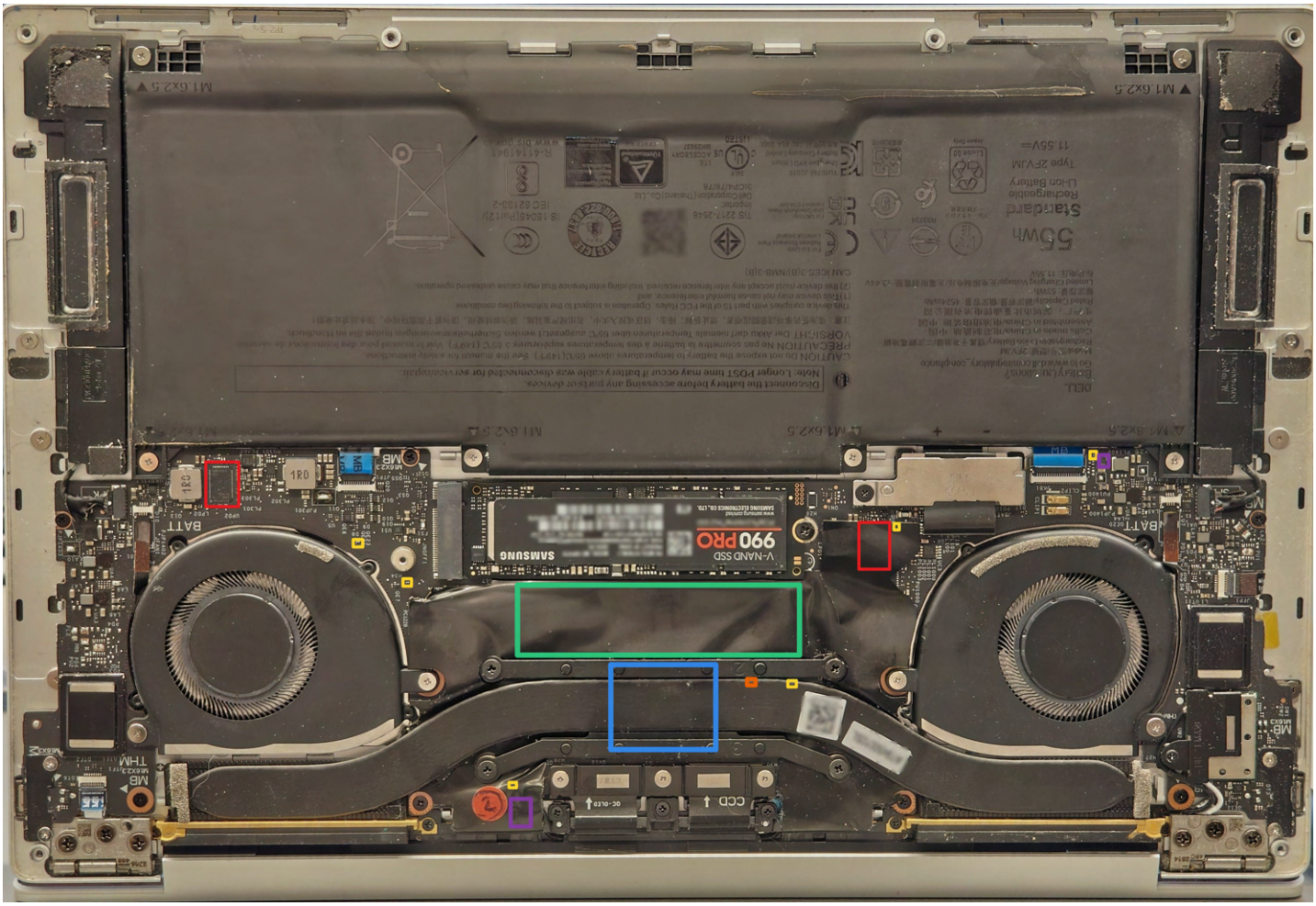
Laptop fans are controlled by EC. Windows driver feeds readings from 7 thermistors to EC over internal i2c bus.

This experiment aims to identify and map physical location of thermistors to their function for user-friendly labels.

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PCBA Overview

Rear Side

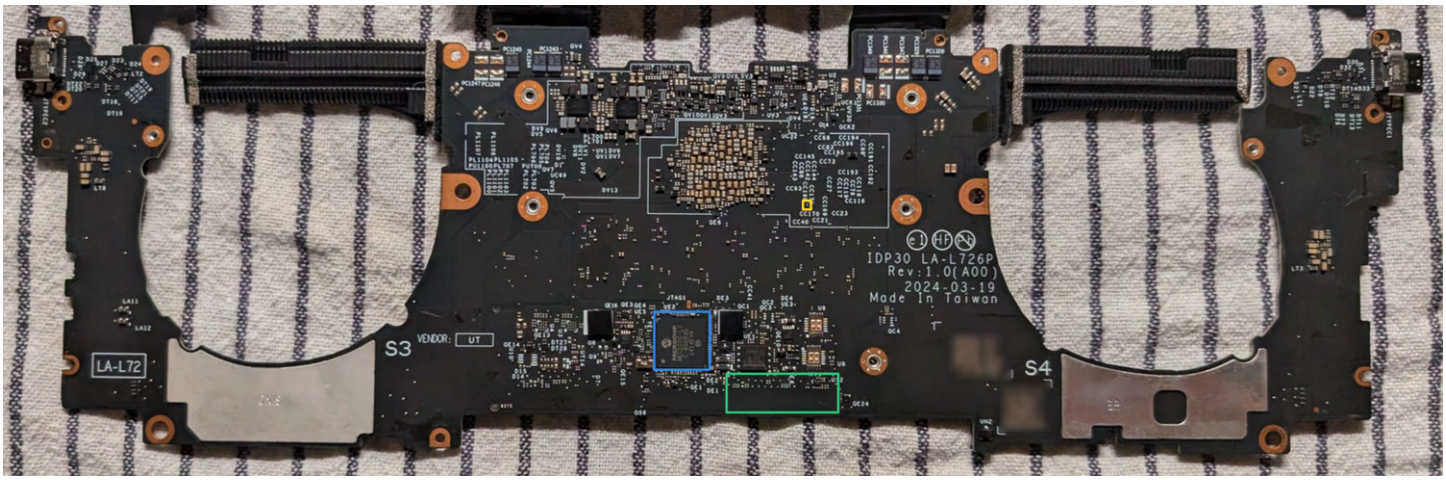


Above picture highlights identified thermistors and key components on the motherboard.

- USB Charging Controller
- SOC
- LPDDR5 RAM
- PMIC/Power IC
- Thermistor (visible PCB side)
- Thermistor (opposite PCB side)

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Front Side



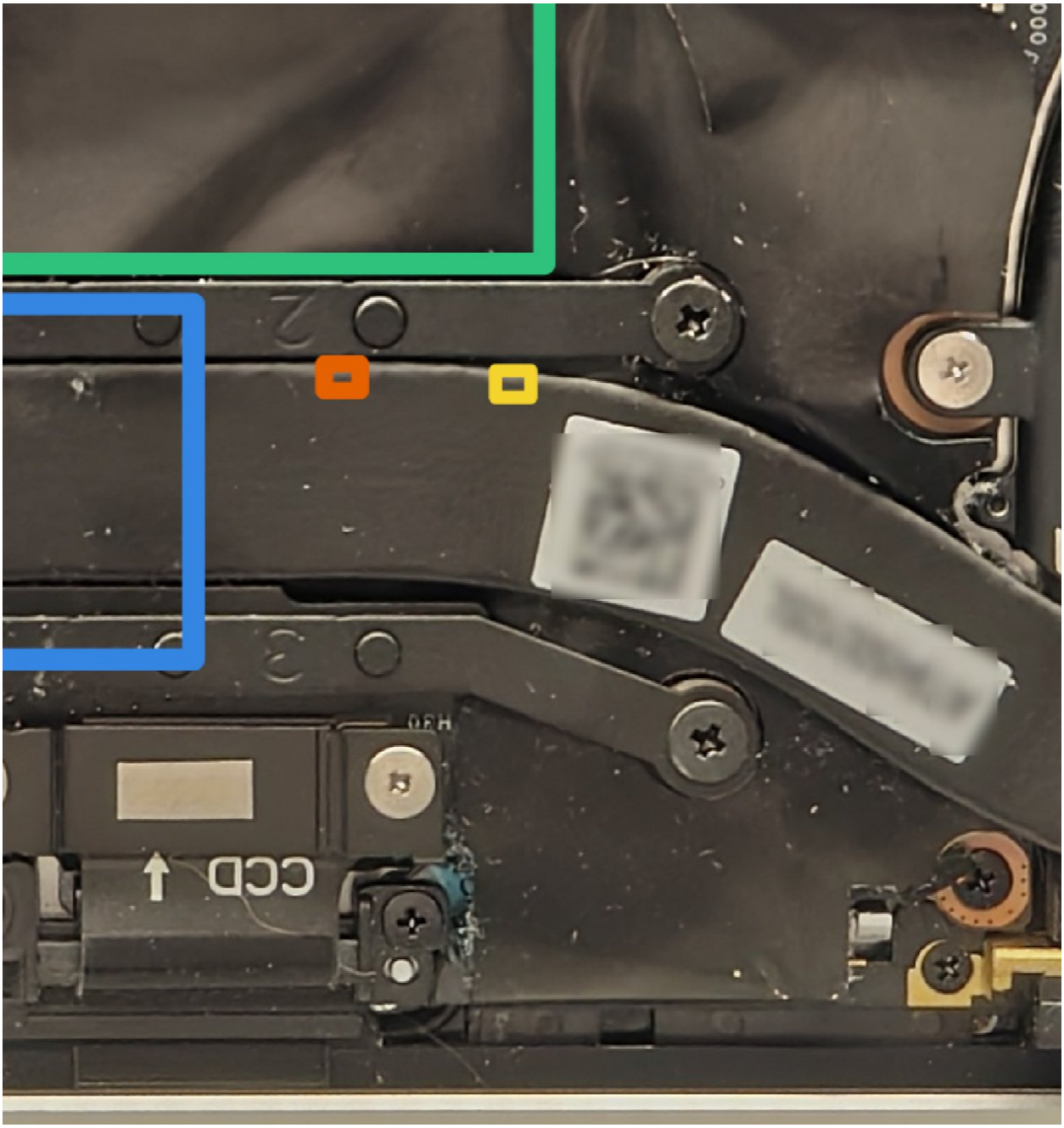
- (Footprint for) Debug Connector
- Embedded Controller (EC)
- Thermistor (visible PCB side)

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SYS_THERM1

Successfully identified in proximity of LPDDR5 RAM. Thermistor (indicated in yellow) cannot be exposed with computer running. To validate, hot air gun was pointed at lifted plastic cover of the motherboard. Reading of the ADC channel 0x4a increased by 14C within 10 seconds, with insignificant increases on other channels.

Due to its position, it can be labeled as `lpddr5_temp` with high confidence.

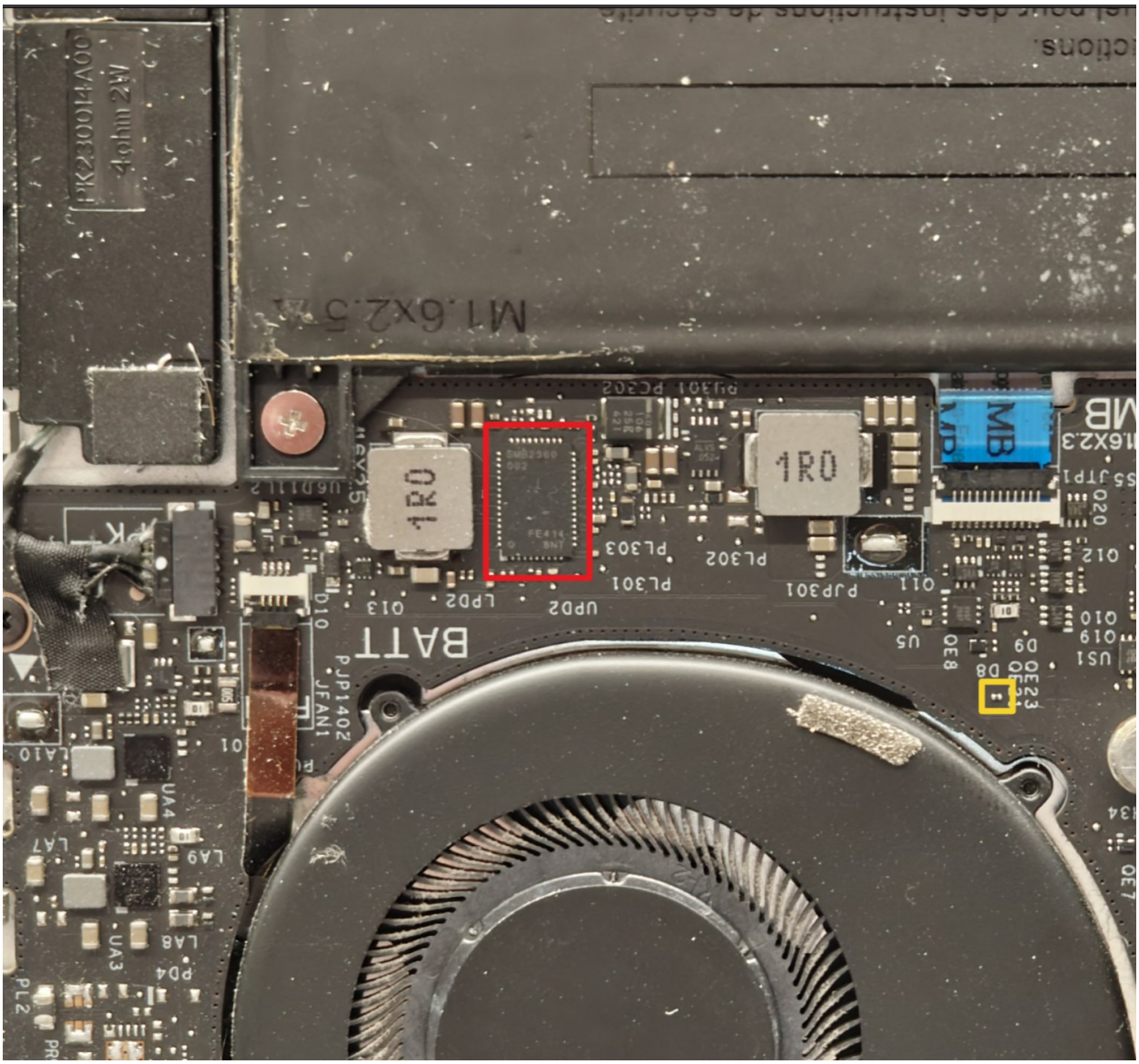


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SYS_THERM2

Successfully identified roughly half way between left-hand-side USB charging controller and SSD connector. Thermistor is exposed with computer running. To validate, soldering iron tip was hovered in close proximity of the thermistor. Reading of the ADC channel 0x4b increased by 27.8C within 10 seconds, with insignificant increases on other channels.

Due to its position, it is difficult to assign this channel to left USB charger.

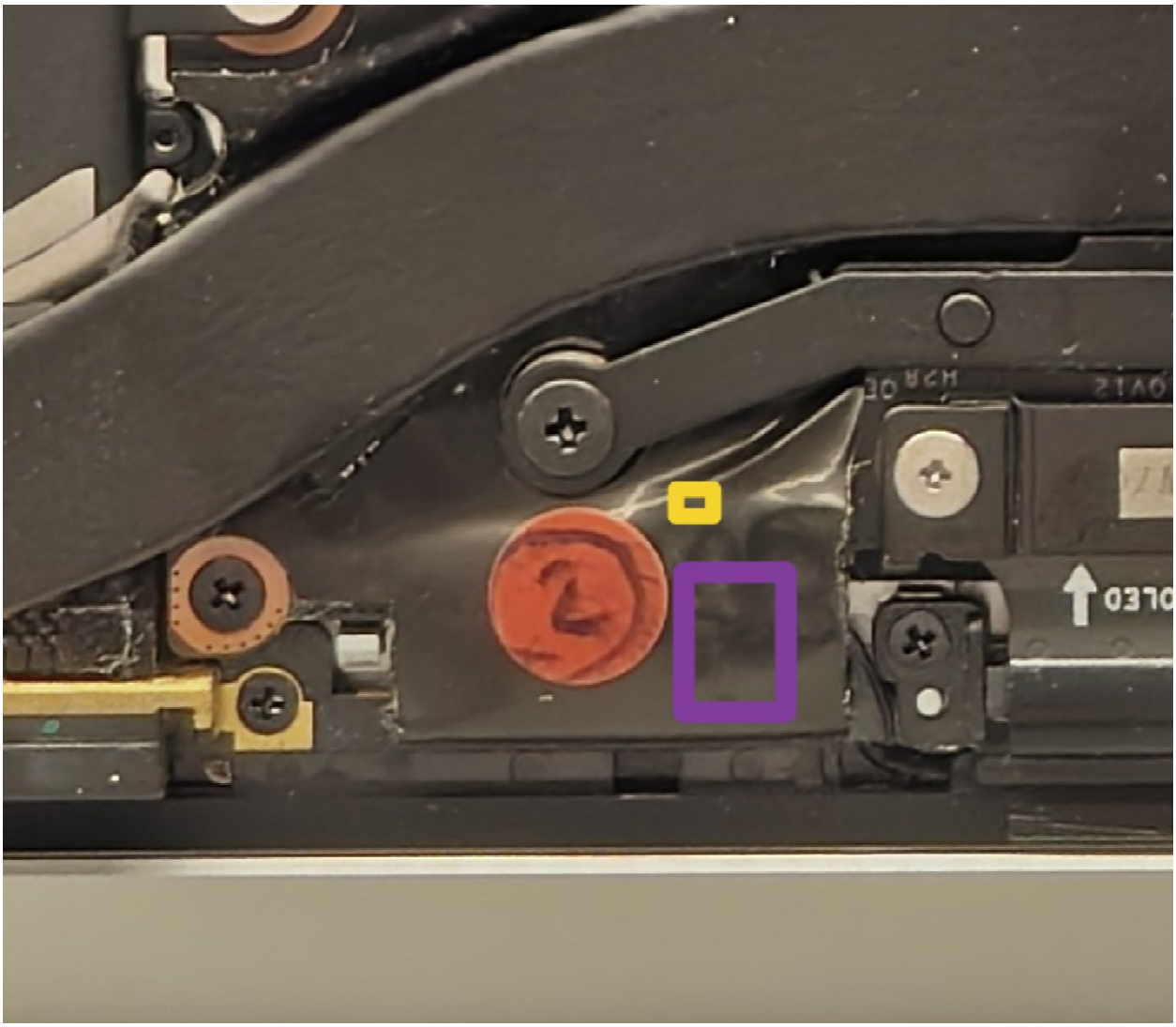


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SYS_THERM3

Successfully identified near main buck power converter, Qualcomm PMC1010H. Thermistor is not exposed with computer running. To validate, hot air gun was pointed at small opening in the plastic cover of the motherboard. Reading of the ADC channel 0x44 increased by 7C within 10 seconds, with insignificant increases on other channels.

Due to its position, it can be labeled as `vph_pwr_temp` with high confidence.

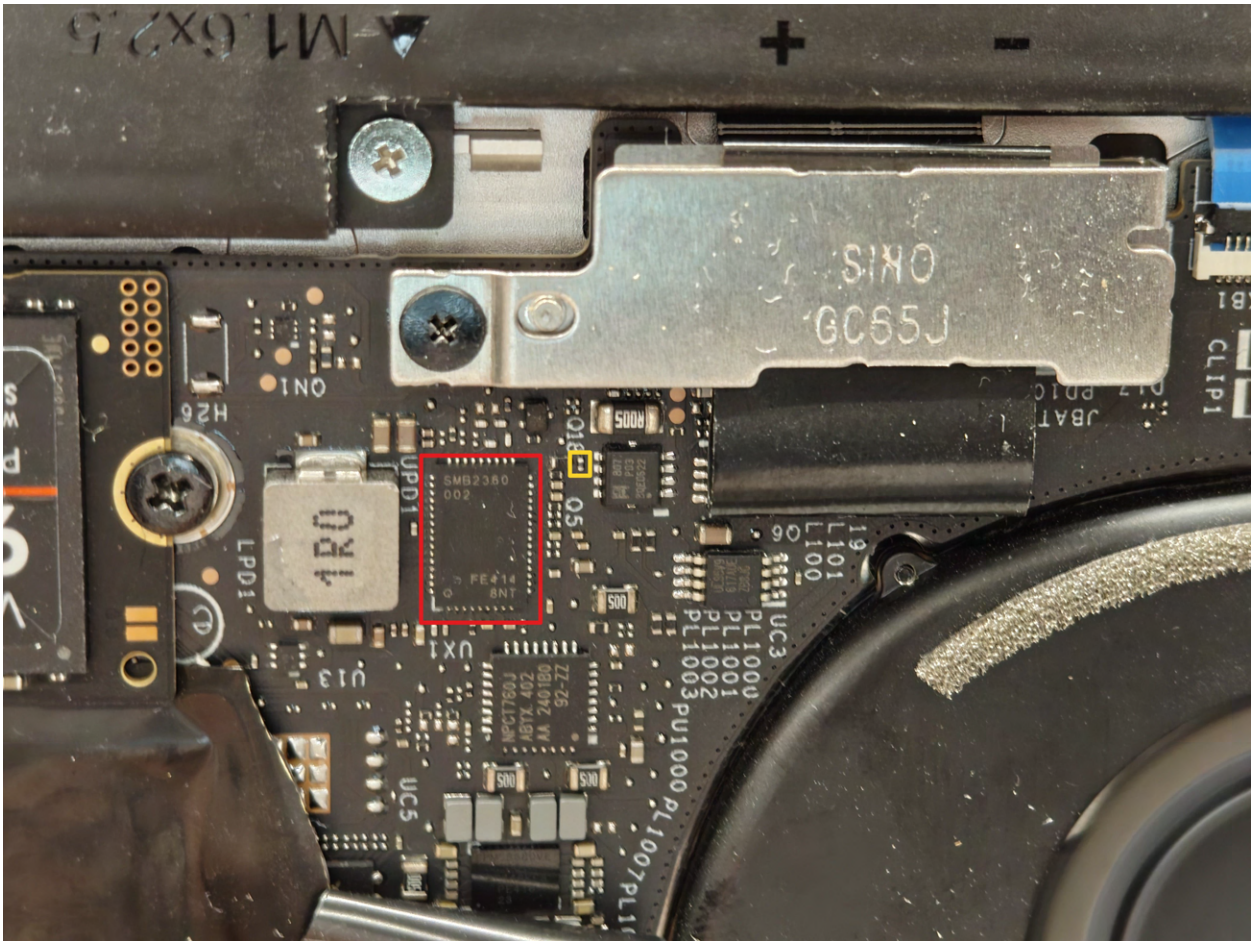


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SYS_THERM4

Successfully identified right next to right-hand-side USB charging controller. Thermistor is exposed with computer running. To validate, soldering iron tip was hovered in close proximity of the thermistor. Reading of the ADC channel 0x45 increased by 53.5C within 10 seconds, with insignificant increases on other channels.

Due to its position, it can be labeled as `usb_charger_right_therm` with high confidence.

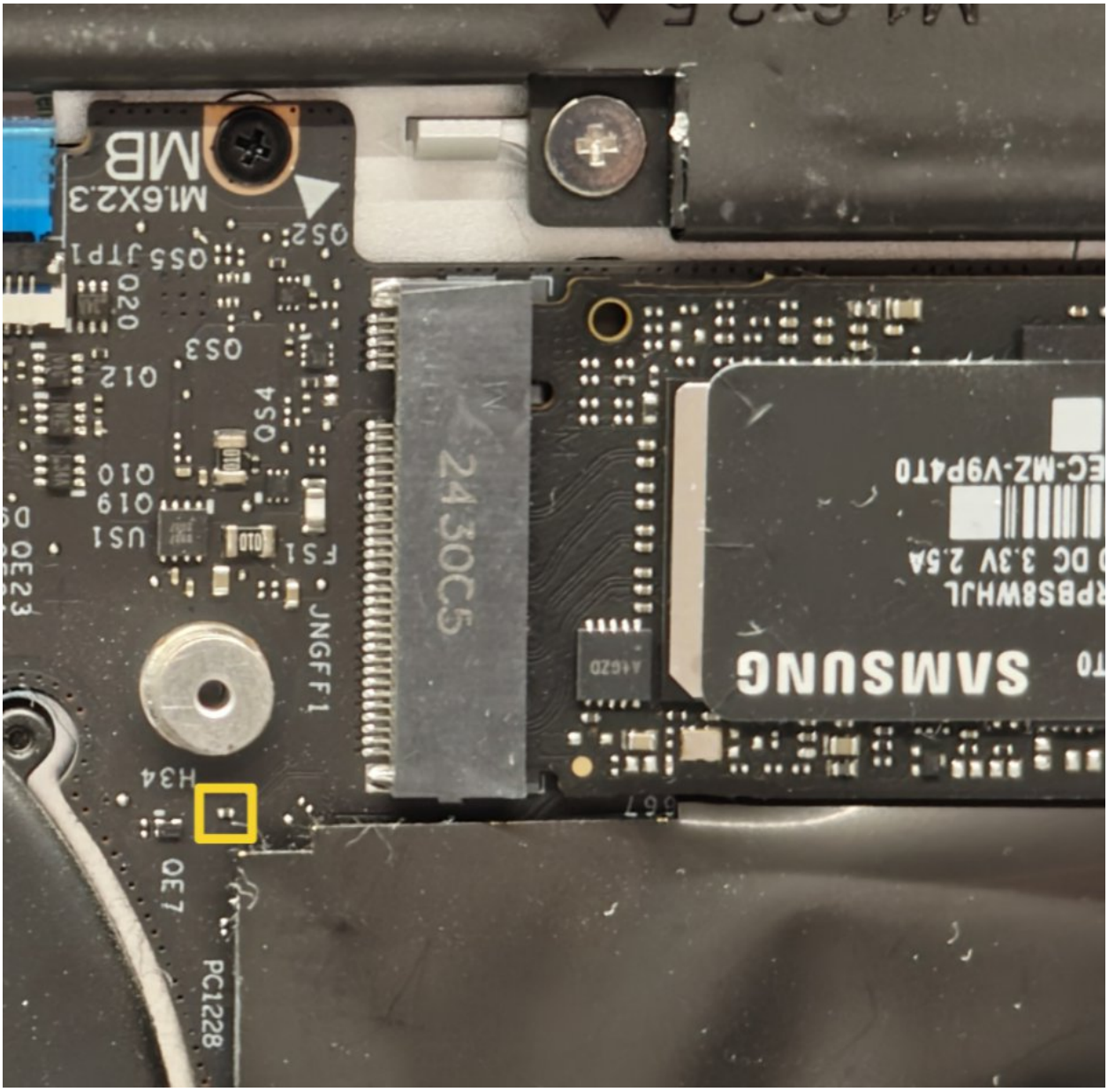


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SYS_THERM5

Successfully identified in close proximity to SSD connector. Thermistor is exposed with computer running. To validate, soldering iron tip was hovered in close proximity of the thermistor. Reading of the ADC channel 0x46 increased by 36.7C within 10 seconds, with insignificant increases on other channels.

Due to its position, it can be labeled as `ssd_conn_therm` with high confidence.

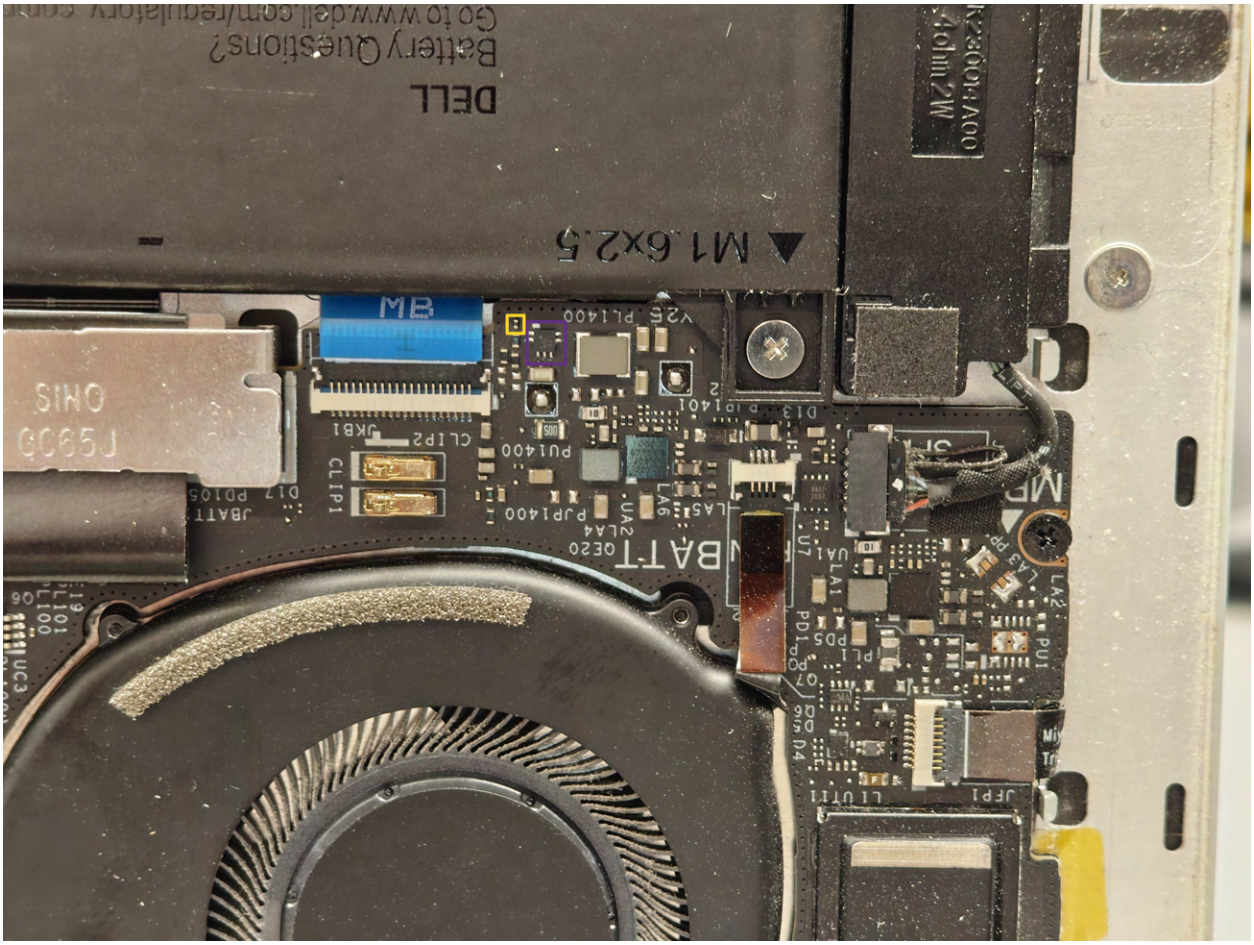


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SYS_THERM6

Successfully identified near a DC-DC buck converted, Silergy SY8843QWC. Thermistor is exposed with computer running. To validate, soldering iron tip was hovered in close proximity of the thermistor. Reading of the ADC channel 0x47 increased by 6.4C within 10 seconds, with insignificant increases on other channels.

Particular buck is supplying `vddar`, `vddat` to right-hand-side Parade PS8830 USB4 retimer.



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SYS_THERM7

Successfully identified on the opposite side of the motherboard. Thermistor is not exposed with computer running, as device must be completed disassembled for thermistor to be accessible. No explicit validation of channel mapping could be performed.

Due to this being the last thermistor from the list, it can be labeled as `keyboard_therm` with high confidence.

